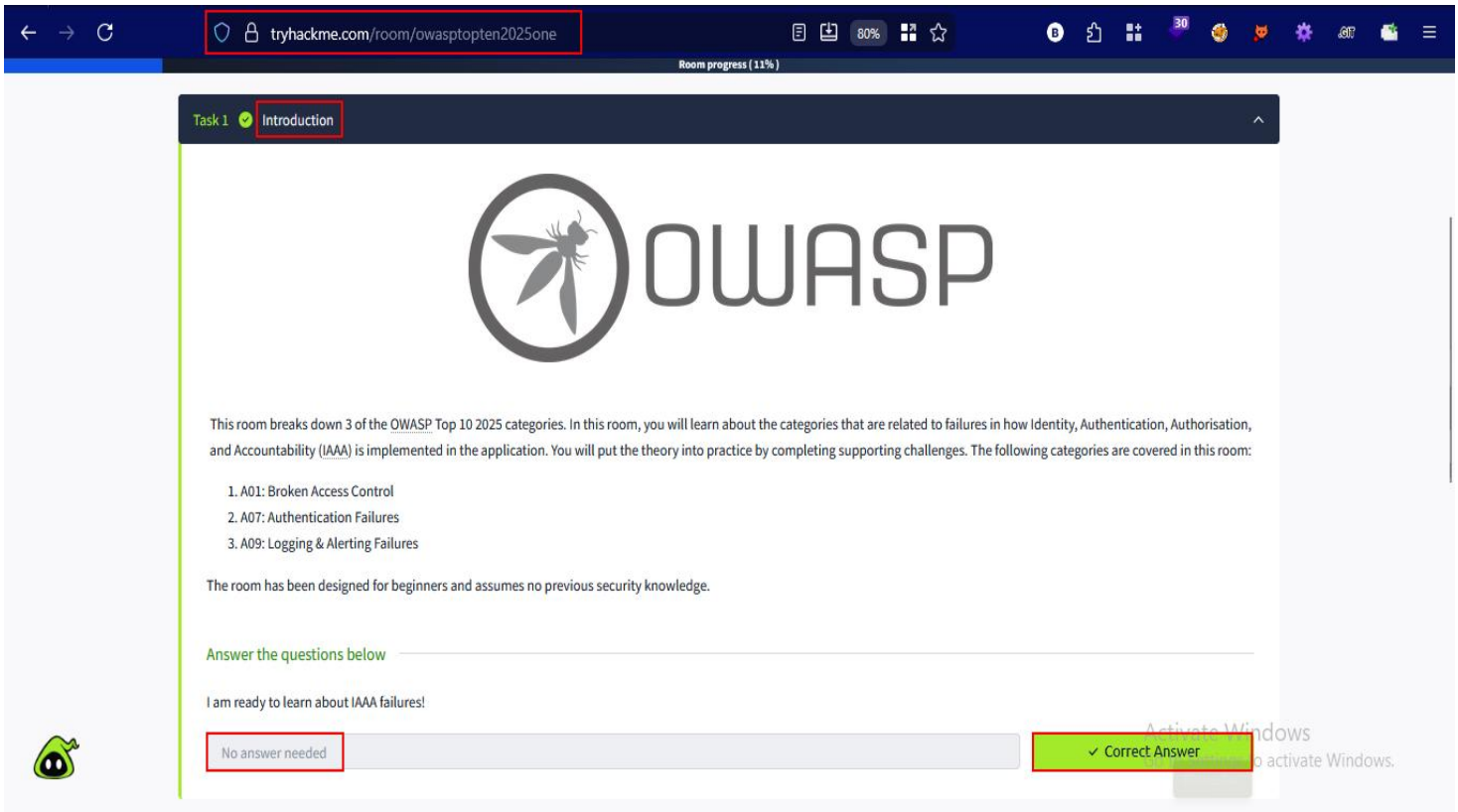
OWASP Top 10 – Module Completion Report

OWASP Top 10 2025 – IAAA Failures

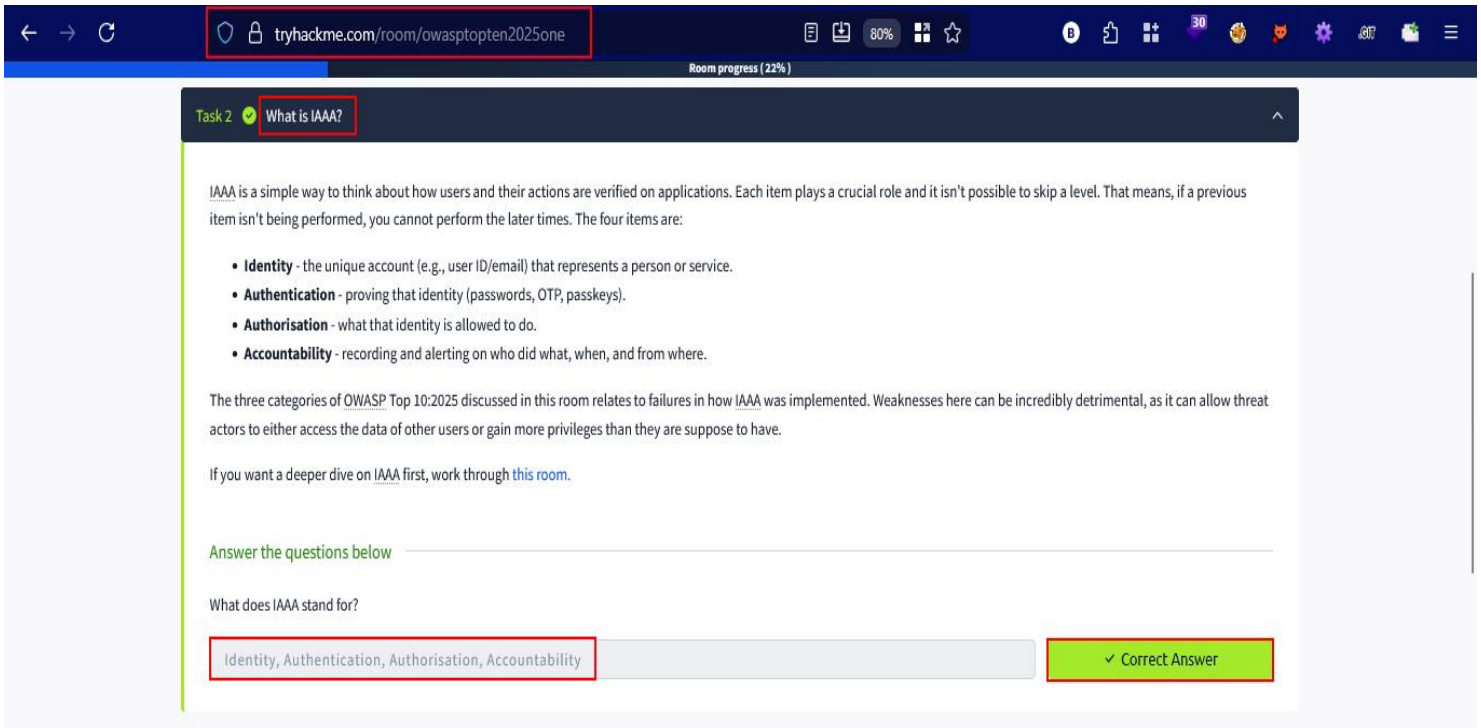
Task 1: Introduction

Description:
This task introduced the fundamentals of the IAAA model and explained its importance in modern web application security. It provided the foundation required to understand identity, authentication, authorisation, and accountability failures.



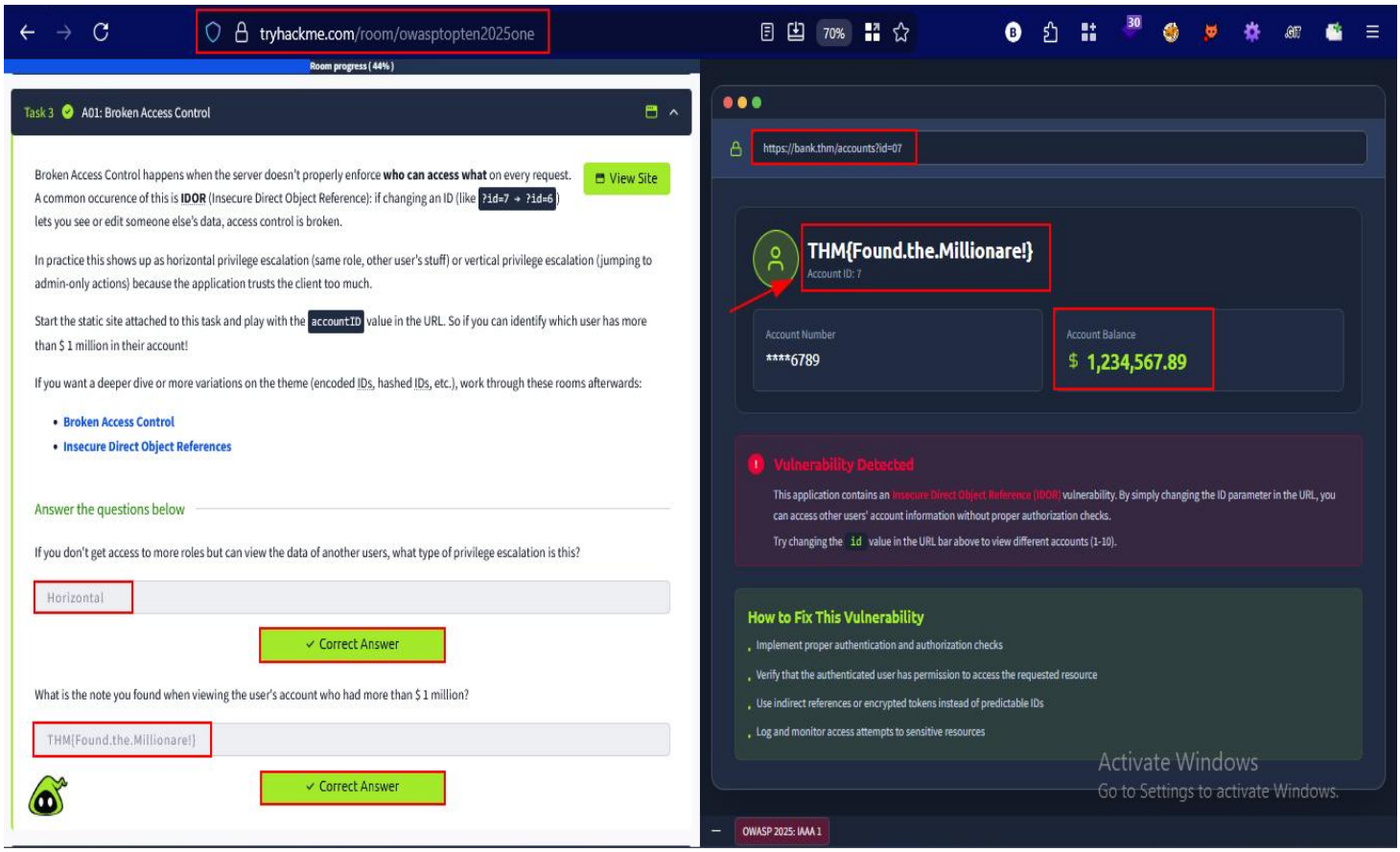
Task 2: What is IAAA?

Description:
This task explained the core components of the IAAA model and how improper implementation can result in serious security weaknesses in web applications.



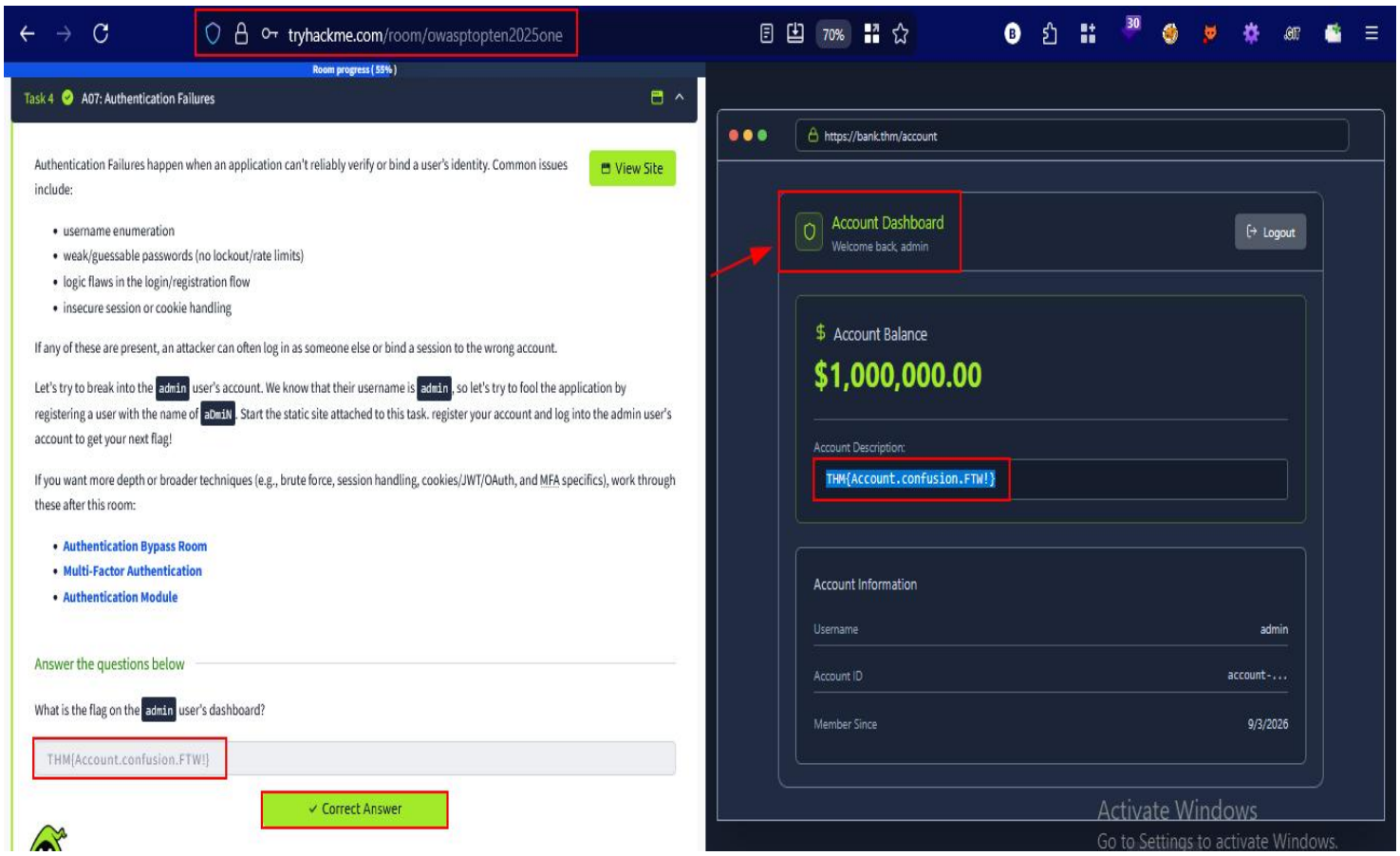
Task 3: A01 – Broken Access Control

Description:
This section focused on access control weaknesses, showing how improper authorization checks can allow users to access resources or functionality beyond their intended privileges.



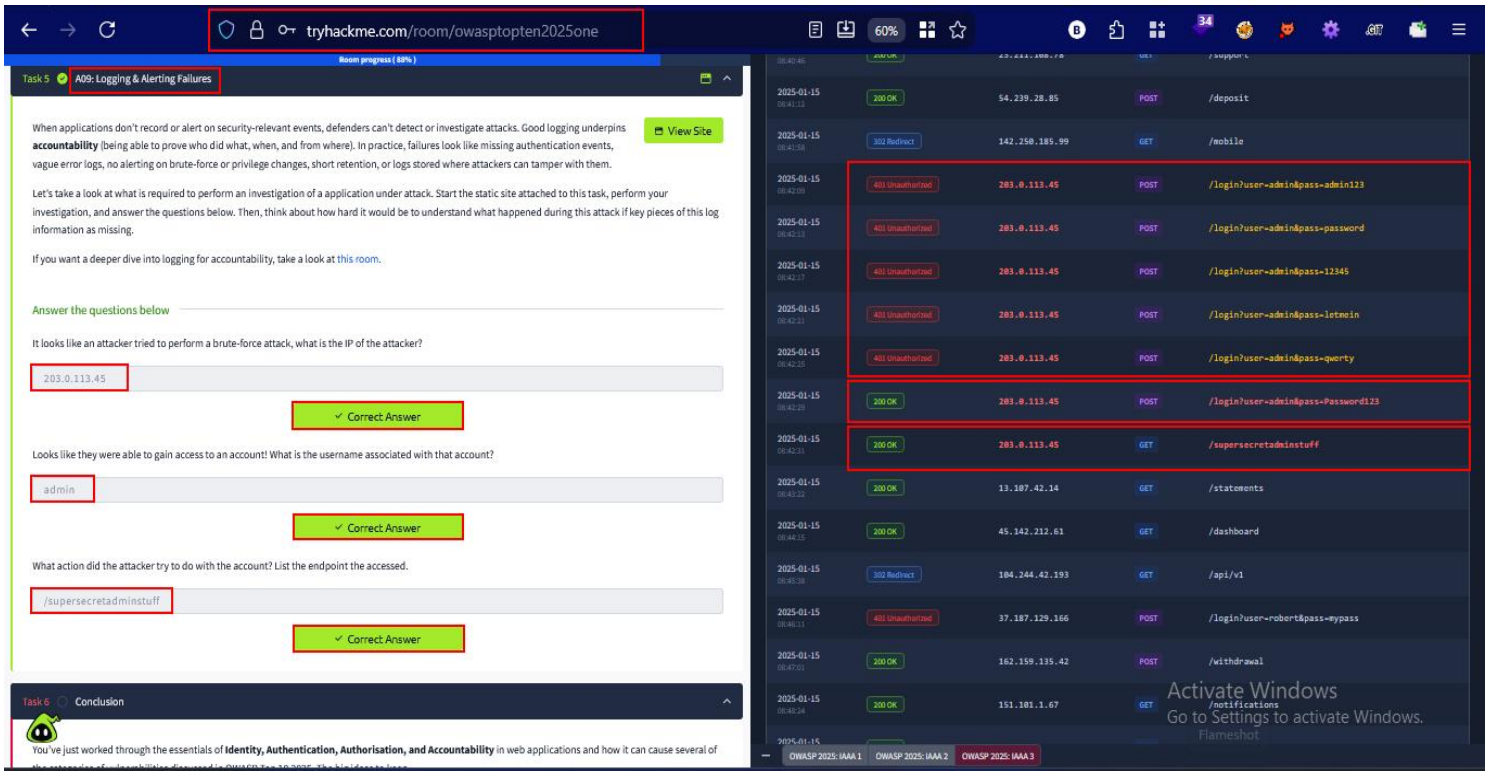
Task 4: A07 – Authentication Failures

Description:
This task covered common authentication weaknesses such as poor session handling, insecure login flows, and weak credential protection mechanisms.



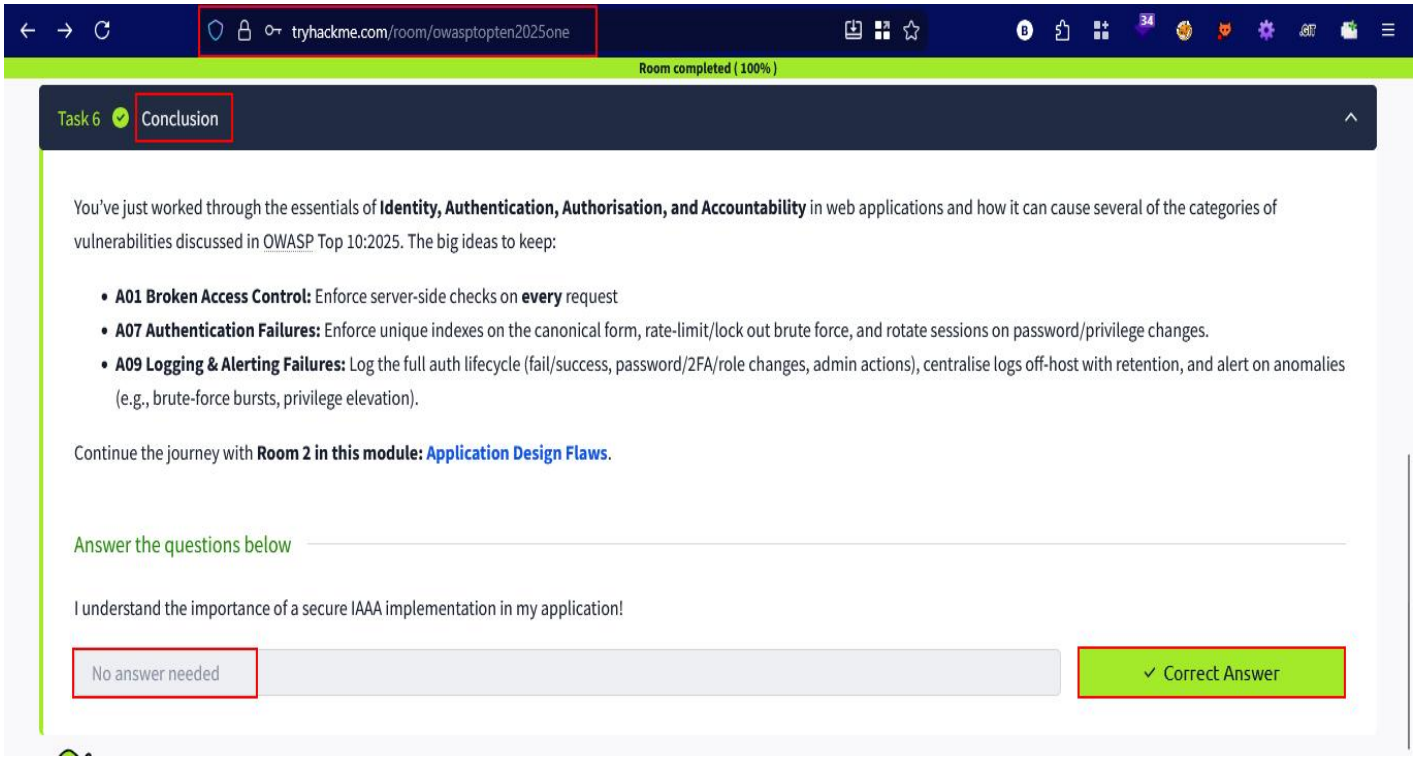
Task 5: A09 – Logging & Alerting Failures

Description:
This section explained how insufficient logging, poor monitoring, and lack of alerting can prevent timely detection of malicious activities or suspicious behavior.



Task 6: Conclusion

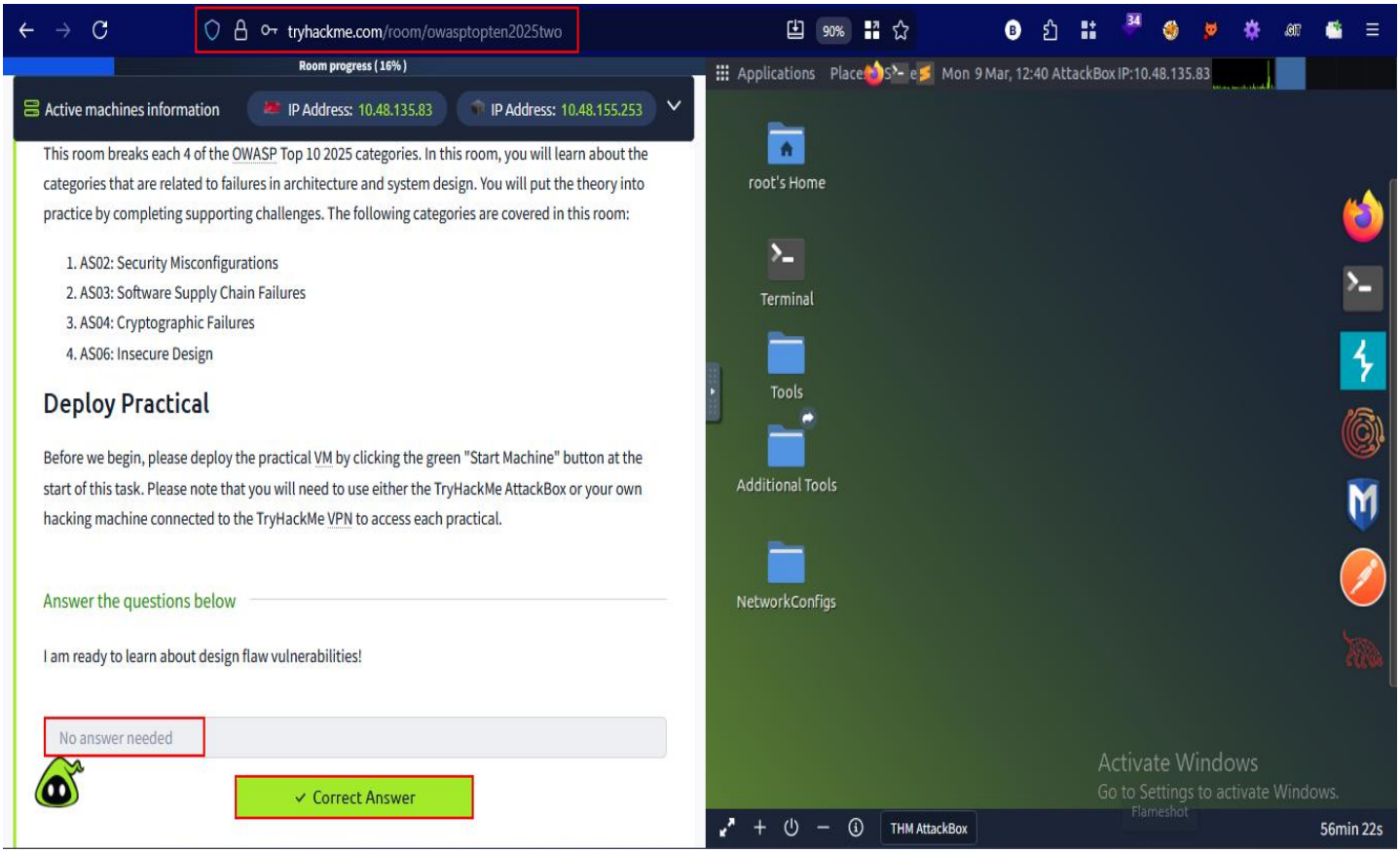
Description:
This module concluded with a summary of IAAA-related security failures and reinforced the importance of implementing secure identity and access management practices.



OWASP Top 10 – Application Design Flaws

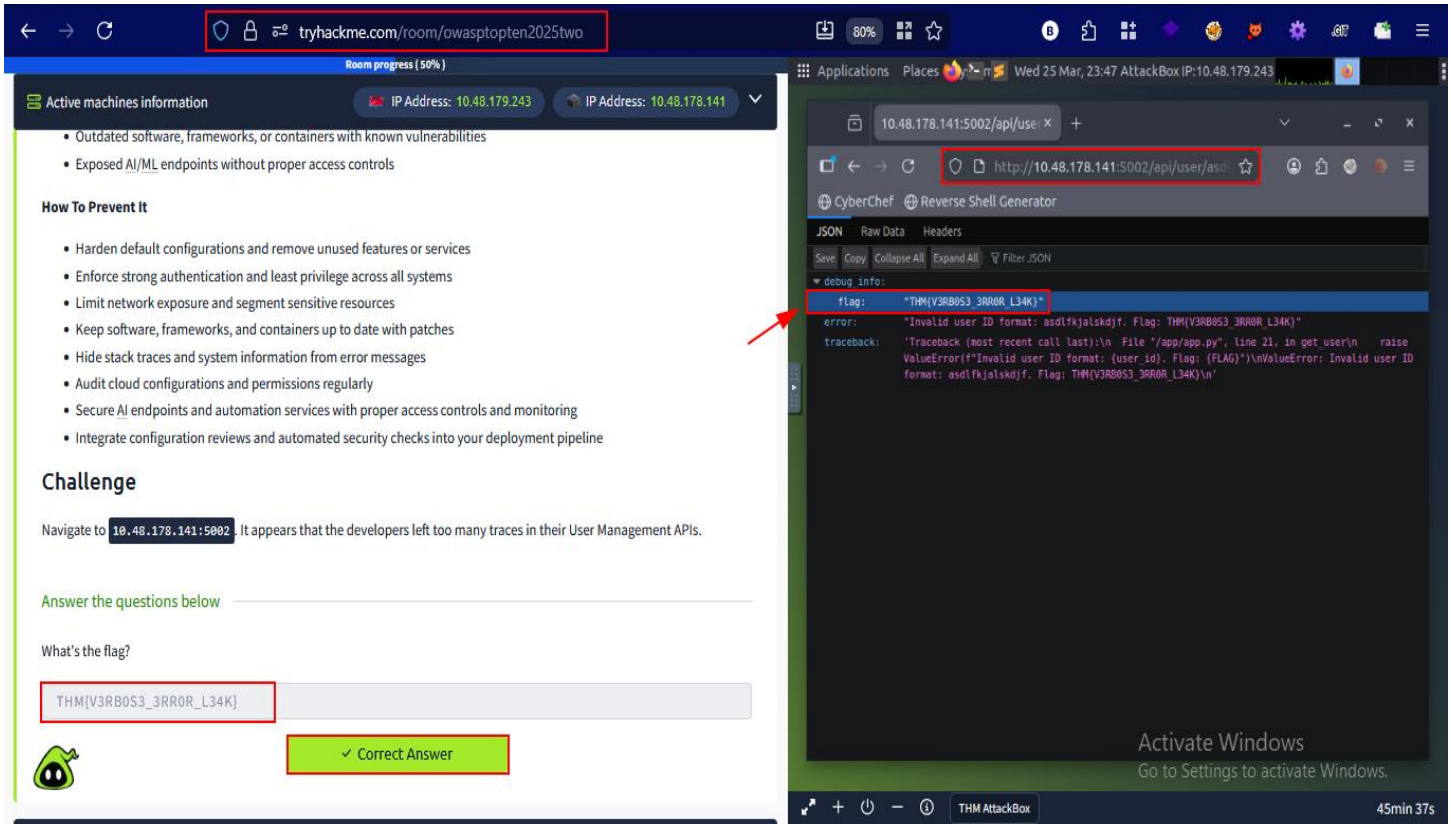
Task 1: Introduction

Description:
This task introduced application design flaws and explained how insecure design decisions can create security weaknesses in web applications. It established the importance of secure planning, architecture, and defensive development practices.



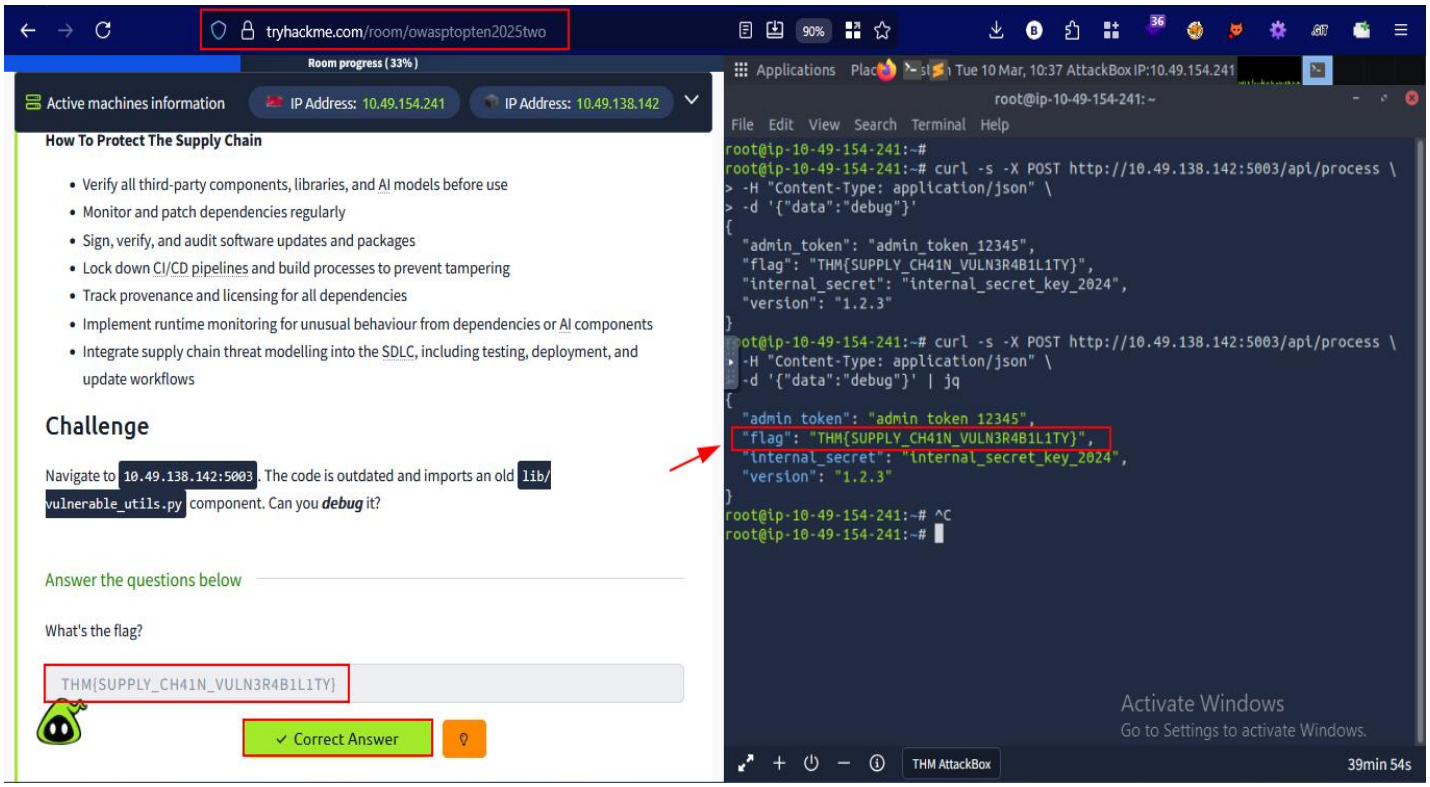
Task 2: AS02 – Security Misconfigurations

Description:
This task explained how incorrect configurations, exposed debug settings, unnecessary services, and insecure defaults can weaken application security and increase attack surface.



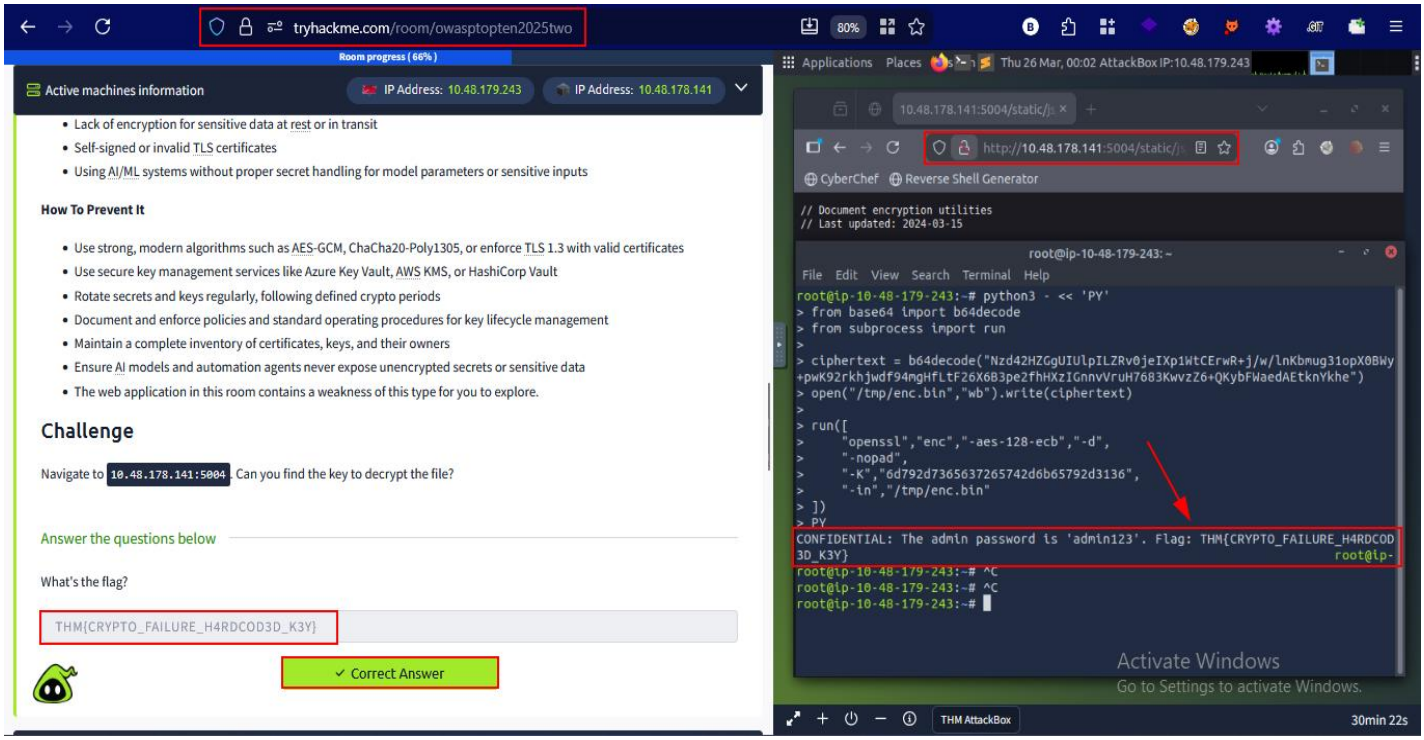
Task 3: AS03 – Software Supply Chain Failures

Description:
This section focused on the risks associated with insecure third-party dependencies, compromised packages, and weak software supply chain security controls.



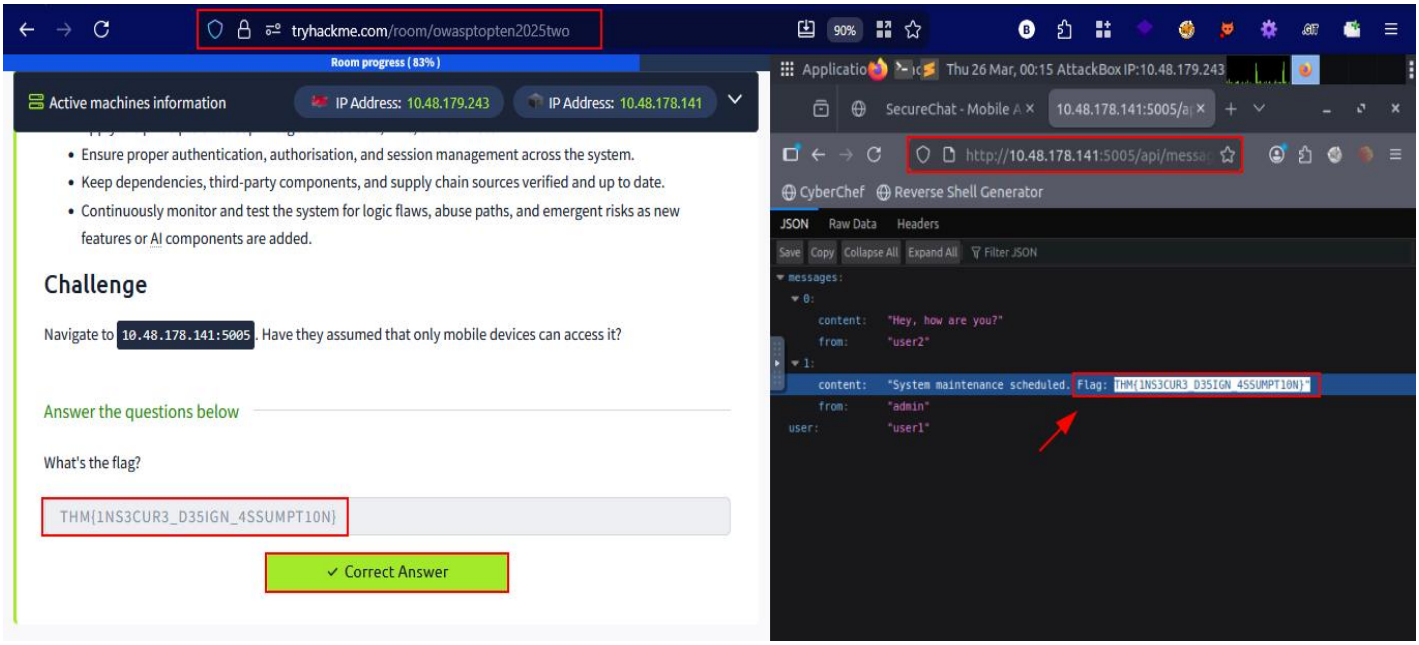
Task 4: AS04 – Cryptographic Failures

Description:
This task covered weaknesses caused by poor encryption practices, insecure key handling, weak algorithms, and improper protection of sensitive information.



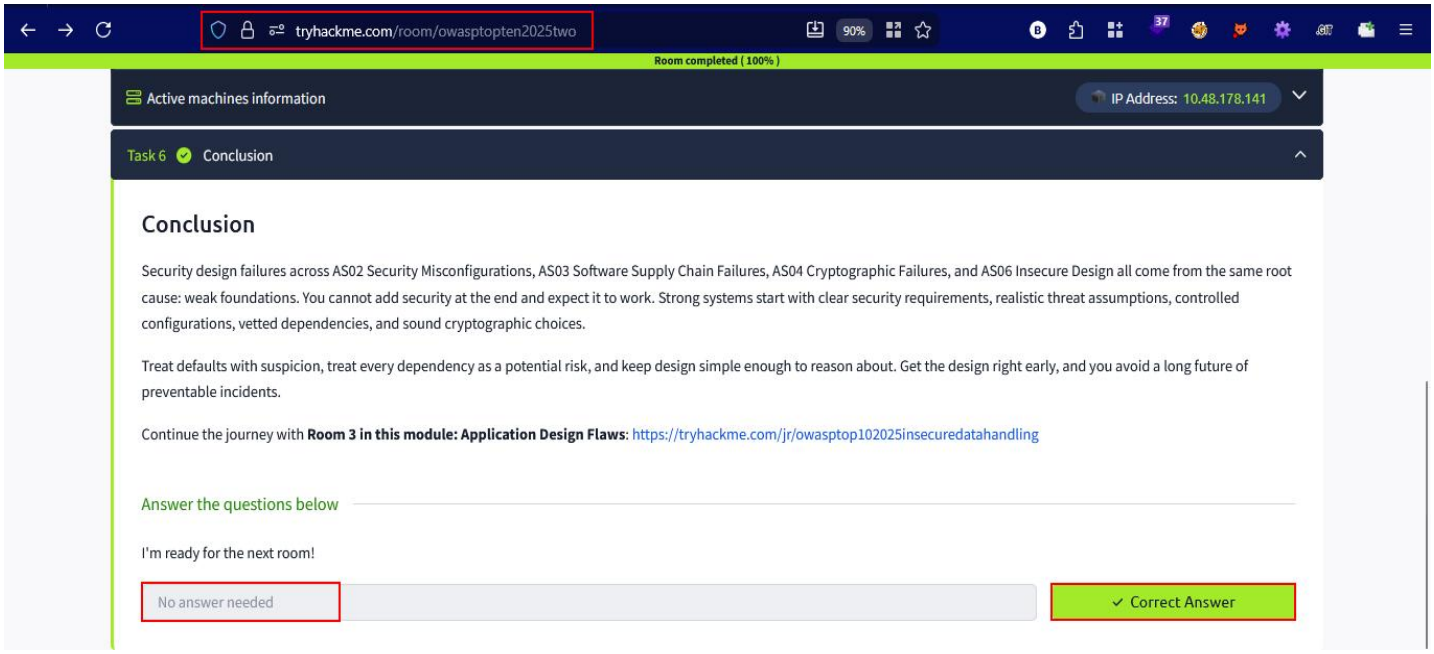
Task 5: AS06 – Insecure Design

Description:
This section explained how missing threat modelling, weak business logic, and poor security-by-design principles can result in vulnerabilities at the application architecture level.



Task 6: Conclusion

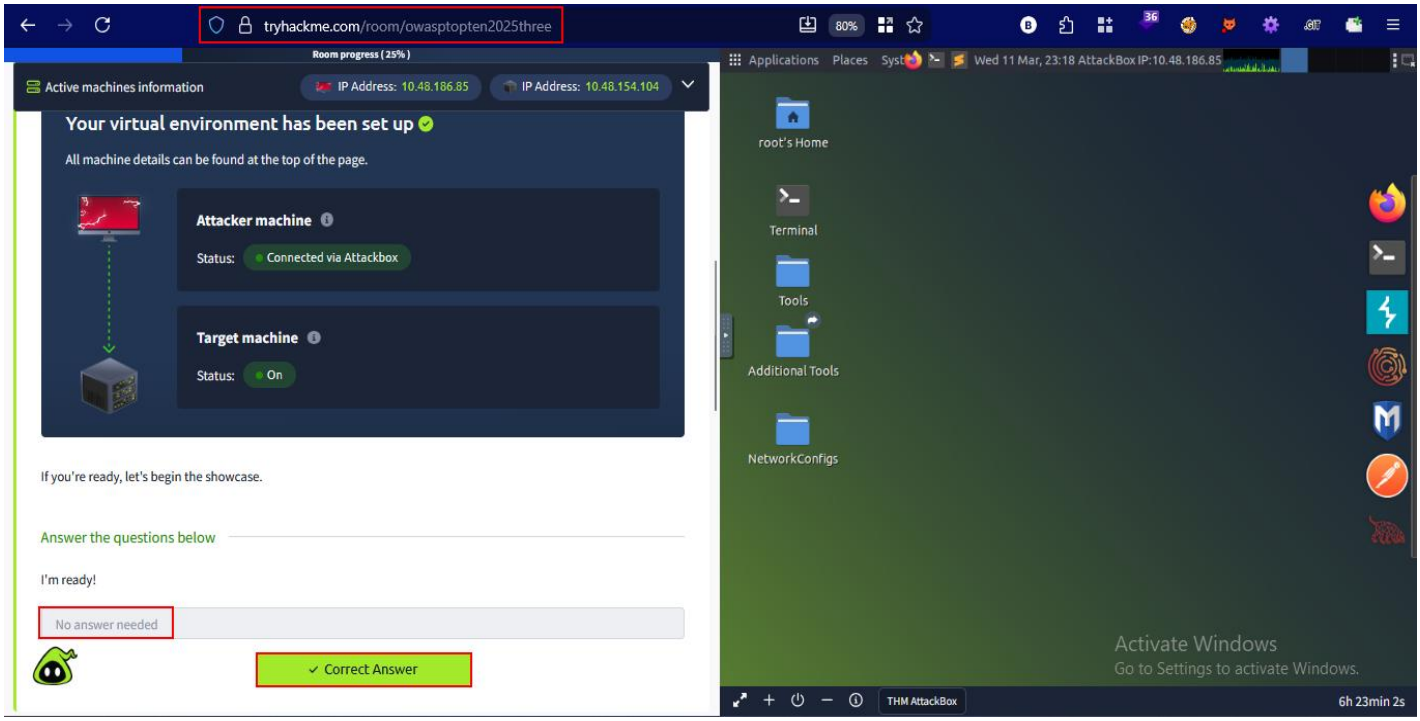
Description:
This module concluded by reinforcing the importance of secure application architecture, strong cryptographic controls, safe configurations, and trusted software development practices.



OWASP Top 10 – Insecure Data Handling

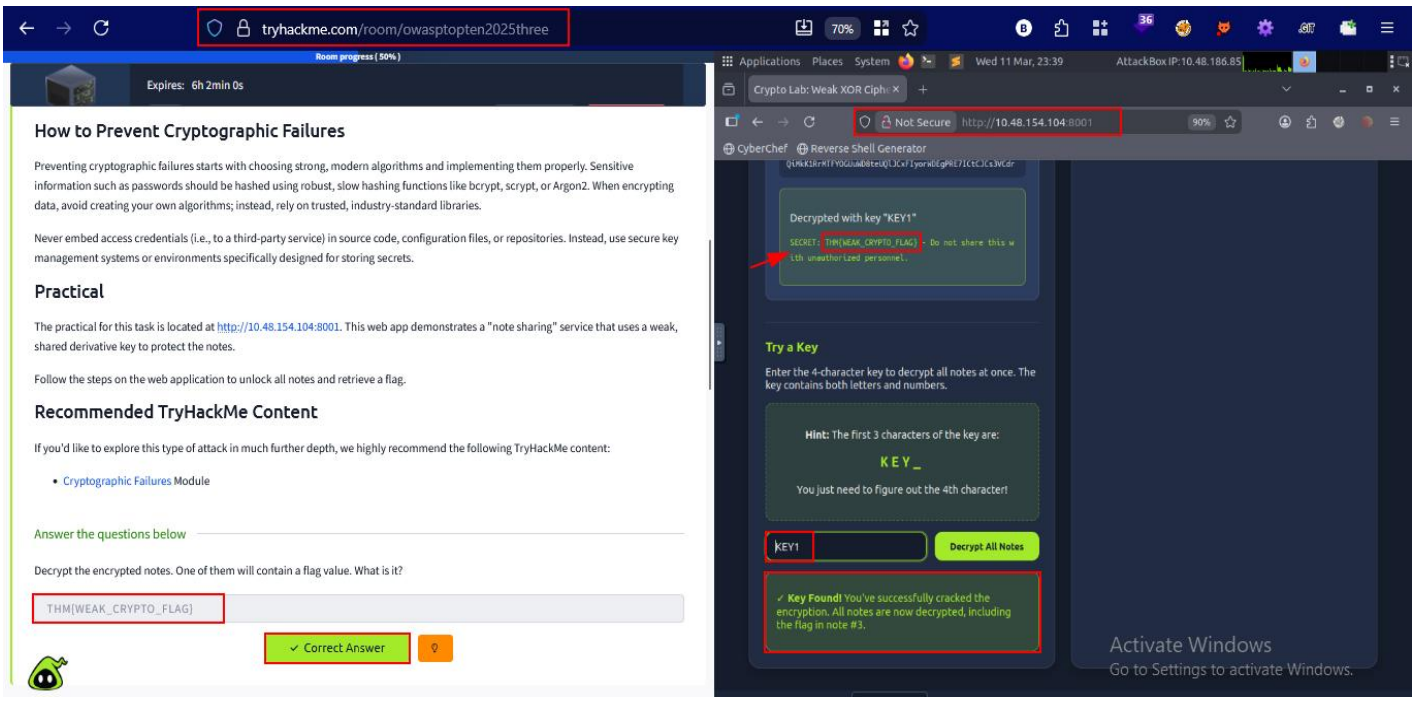
Task 1: Introduction

Description:
This task introduced insecure data handling issues and explained how improper processing, storage, and validation of data can lead to serious application security risks.



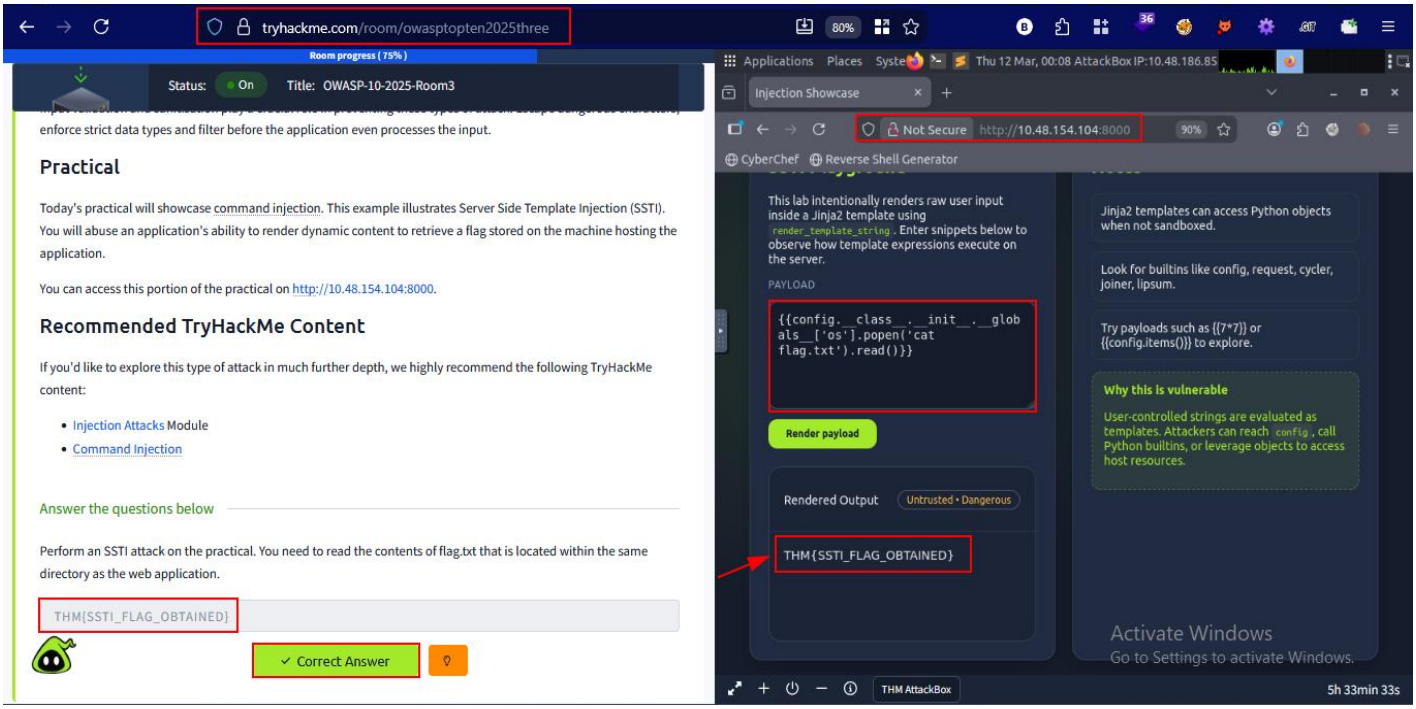
Task 2: A04 – Cryptographic Failures

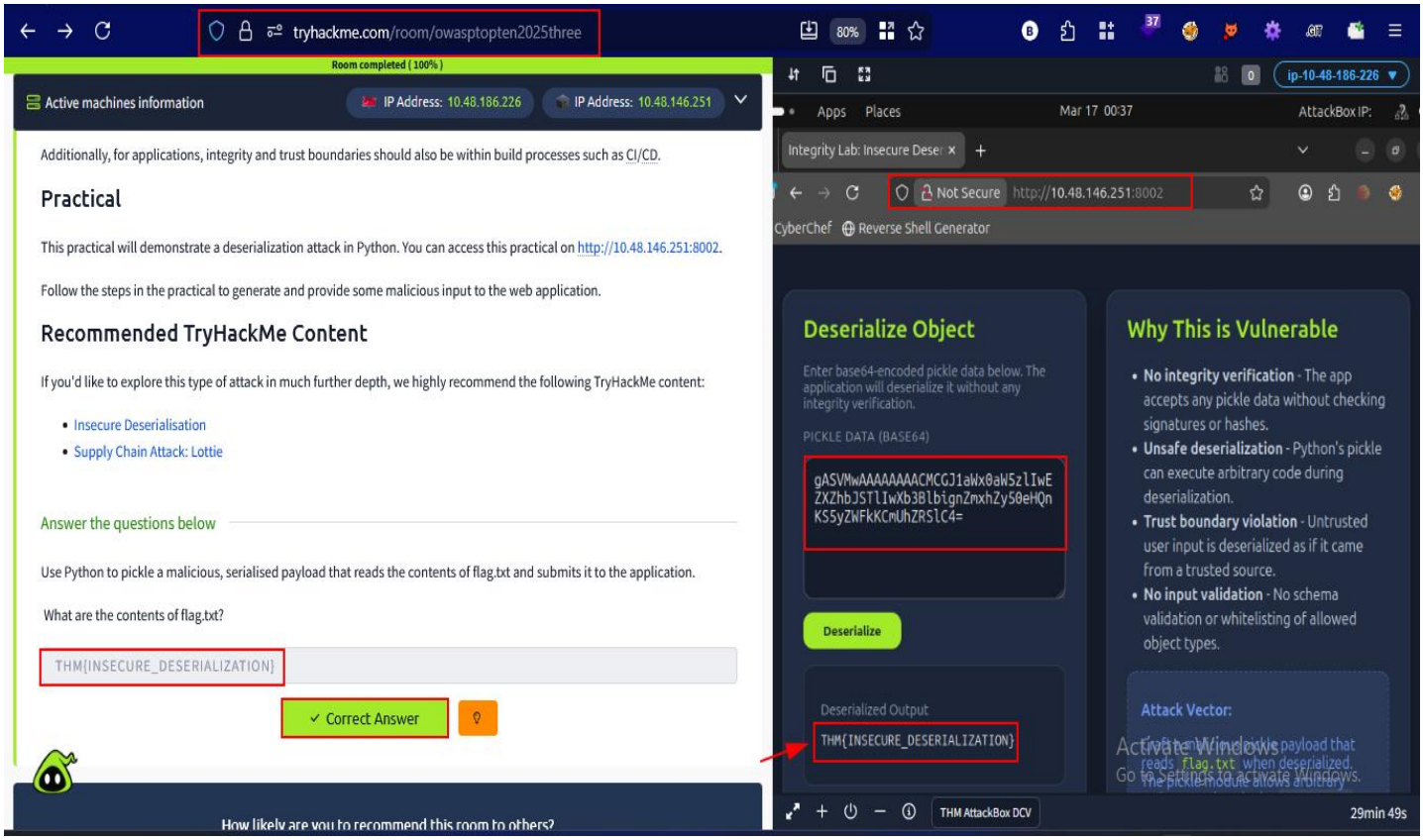
Description:
This task explained how weak encryption, insecure transmission, and poor handling of secrets can expose confidential data and reduce overall application security.



Task 3: A05 – Injection

Description:
This section demonstrated how unsanitised or untrusted input can be abused by attackers to manipulate application behavior through injection-based attacks.



Task 4: A08 – Software or Data Integrity Failures

Description:
This task focused on failures related to trust, integrity verification, insecure updates, and the unsafe use of code or data from untrusted sources.